25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Limits, Continuity & Introductory Derivatives Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?

A $2 + 3i$

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

2 MCQ

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

A 0

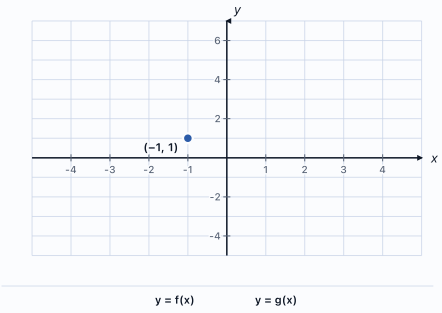
B 2

C -1

D 3

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?



A (1, 0) and (3, 0)

B (0, 1) and (0, 3)

C (-1, 0) and (-3, 0)

D (1, 0) and (-3, 0)

4 MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

A [3, 4]

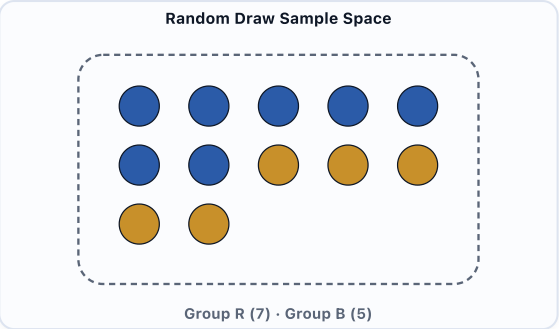
B [5, 8]

C [-3, -4]

D [4, 3]

5 MCQ

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?



A $\frac{3}{5}$

B $\frac{2}{5}$

C $\frac{1}{2}$

D $\frac{3}{4}$

6 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

A 14

B 17

C 8

D 11

7 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

A 96

B 192

C 98

D -96

8 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

A 0

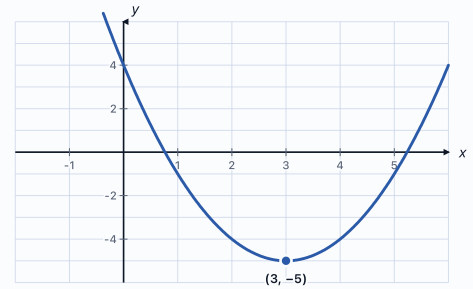
B -2

C 2

D 1

9 MCQ

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph has a vertex at $(3, -5)$. Determine the value of the parameter k.



A 9

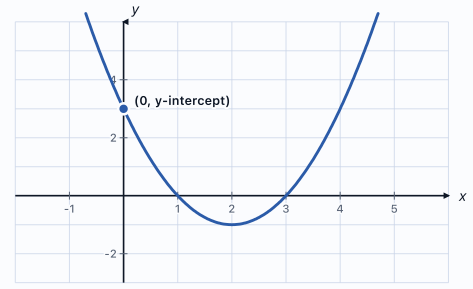
B -4

C 4

D 1

10 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What is the y-intercept of the graph of this function?



A -3

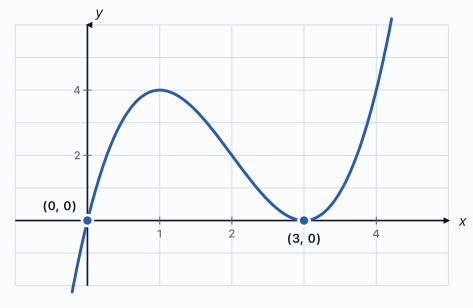
B 4

C -4

D 3

11 MCQ

Consider the function $f(x) = x^3 - 6x^2 + 9x$. Find the area of the region bounded by the graph of $f(x)$ and the x-axis.



A 10

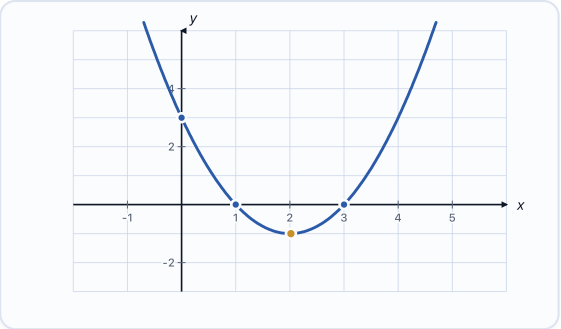
B 27/2

C 27/4

D 9

12 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. What is the y-intercept of this parabola?



A -3

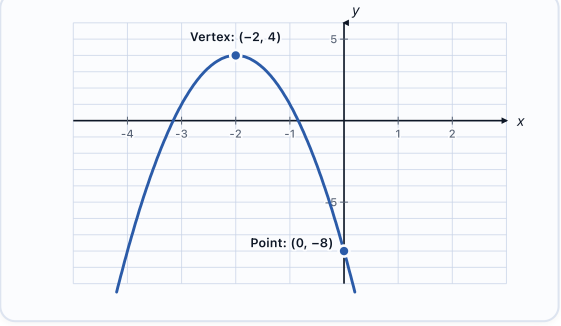
B 0

C -4

D 3

13 MCQ

Consider the function $g(x) = ax^2 + bx + c$. The graph of this quadratic function is shown below. The vertex of the parabola is at $(-2, 4)$. The parabola passes through the point $(0, -8)$. Find the value of the parameter a.



A 2

B 4

C -4

D -2

14 MCQ

Consider the function $f(x) = x^3 - 6x^2 + 5$. A tangent line to the curve of $f(x)$ is drawn at the point $P(a, f(a))$. This tangent line also intersects the curve at another point $Q(b, f(b))$. If the x-coordinate of the point P is $a = 1$, determine the x-coordinate of the point Q.

A 4

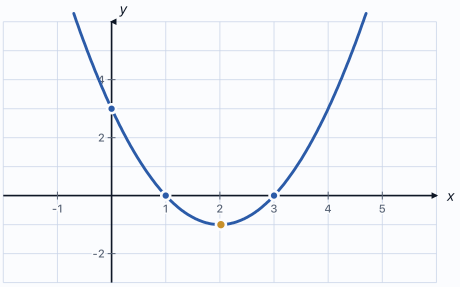
B 6

C 5

D 3

15 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. Find the x-coordinate of the vertex of the parabola.



A 2

B 1

C 3

D -1

16 MCQ

Find $\lim_{x \to -3} (-x^2 - 3x - 3)$.

A -5

B 3

C -3

D -2

17 MCQ

Find $\lim_{x \to -3} (4x^2 + 4x - 5)$.

A 17

B 19

C -19

D 20

18 MCQ

Find $\lim_{x \to 3} (-x - 2)$.

A -4

B -5

C -7

D 5

19 MCQ

Find $\lim_{x \to -5} (2x + 6)$.

A -3

B -4

C -6

D 4

20 MCQ

Find $\lim_{x \to 2} (-2x^2 + 2x - 7)$.

A -11

B 11

C -10

D -13

21 MCQ

Find $\lim_{x \to 5} (3x^2 - 7x - 5)$.

A 36

B 35

C 33

D -35

22 MCQ

Find $\lim_{x \to 4} (-3x^2 - 6x + 2)$.

A -69

B -72

C 70

D -70

23 MCQ

Find $\lim_{x \to 0} (2x^2 + 2x)$.

A -2

B 4

C 1

D 0

24 MCQ

Calculate $\lim_{x \to 3} (-2x^2 + 5x - 5)$.

A -8

B -7

C -10

D 8

25 MCQ

Find $\lim_{x \to -1} (-4x^2 + x - 1)$.

A -5

B -6

C -8

D 6